Gabriel Espinoza

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Education

George Mason University Fairfax, VA
Master of Science in Computer Science *Expected Dec. 2027*

GPA: 4.0/4.0
Relevant Coursework: Advanced Algorithms, Mathematical Foundations of Computer Science, Computer Communication & Networking, Artificial Intelligence, Computer Graphics, Mobile Immersive Computing

George Mason University Fairfax, VA
Bachelor of Science in Computer Science *Jan. 2023 – Aug. 2025*

Relevant Coursework: Operating Systems, Data Structures, Analysis of Algorithms, Computer Systems and Programming, Database Concepts, Software Engineering, Natural Language Processing, Data Mining, Probability & Statistics, Numerical Analysis, Secure Programming and Systems, Foundations and Applications of Cybersecurity

Technical Skills

Languages: Python, C, Java, JavaScript, SQL, Bash, HTML/CSS
Frameworks: FastAPI, Node.js, Express.js, React, Tailwind CSS
Developer Tools: VScode, Git, Github Actions, Docker, Prometheus, CI/CD, Jupyter Notebook
Databases: PostgreSQL, MySQL, Redis
Platforms: Linux, macOS, Windows

Projects

Custom Operating System Kernel | *C, Git, Linux, GDB* January 2025 – May 2025

- Built multi-process support for an operating system kernel by implementing core process creation, execution, synchronization, and termination functionality
- Designed process state to outlive threads by decoupling process structs from the thread lifecycle, enabling parents to reliably retrieve exit codes from terminated children via semaphore-based synchronization
- Implemented execv with user-to-kernel argument marshaling and rollback safety, preserving the original address space until the new program loads successfully to prevent process corruption on failure
- Strengthened syscall and fault-handling paths with robust error propagation, graceful user-process termination, and safeguards against kernel panics from invalid user-mode behavior

Tempo | *Python, FastAPI, PostgreSQL, Tortoise ORM, Docker* January 2026 – April 2026

- Engineered a scalable backend for a calendar-based SaaS platform, enabling user authentication, time-anchored event tracking, and context-linked notes to streamline productivity workflows
- Designed and implemented RESTful APIs for core features including user management, event scheduling, and note-event associations, with secure JWT-based authentication and role-based control
- Developed and maintained 15 automated tests across 3 test suites, and integrated a CI pipeline to ensure reliability, catch regressions, and continuously validate core system functionality

Limitr | *Python, FastAPI, Redis, Docker, Nginx, Prometheus, Pytest* February 2026 – April 2026

- Built a distributed rate-limiting reverse proxy using Redis-backed token bucket algorithms
- Implemented fail-closed resilience behavior during Redis outages to prevent unmetered access
- Integrated Prometheus metrics for throughput, rejection rates, Redis latency, and outage duration
- Built a GitHub Actions CI pipeline running 22 automated tests and Locust load testing